

I. THE BACKBONE OF THE DIGITAL WORLD

1.1. Introduction to Networking

In the field of Cyber Security, networks serve as the primary medium through which all digital communication occurs. Every action performed in a digital environment—whether sending emails, accessing websites, streaming media, or launching cyberattacks—relies on network infrastructure. Therefore, understanding how networks function is a fundamental requirement for securing any system.

From a security perspective, the network can be viewed as a **battlefield**, where legitimate users and malicious actors interact. Attackers exploit vulnerabilities in network protocols, configurations, and devices to gain unauthorized access, intercept data, or disrupt services.

To effectively defend against such threats, cybersecurity professionals must understand:

- How data is transmitted between devices
- How communication protocols operate
- Where vulnerabilities are likely to occur

This section introduces the foundational concepts of networking, which form the basis for more advanced security techniques.

1.2. What is a Network?

A network is defined as a collection of interconnected devices that communicate with each other to share data and resources. These devices may include computers, servers, routers, switches, and other networking equipment.

Networks can be categorized based on their scale and coverage:

- **LAN (Local Area Network):** A network that covers a small geographic area such as a home, office, or campus. Typically uses Ethernet or Wi-Fi technology.
- **WAN (Wide Area Network):** A network that spans large geographical areas. The Internet is the largest example of a WAN, connecting millions of networks worldwide.

In cybersecurity, both LANs and WANs must be secured, as attackers can target internal systems (LAN) or external connections (WAN).

1.3. Key Network Hardware

Network infrastructure relies on several essential hardware components:

- **Routers:** Devices responsible for directing data packets between different networks. They analyze IP addresses and determine the most efficient path for data transmission. In security, routers often include basic firewall features.
- **Switches:** Operate within a local network to connect multiple devices. They use MAC addresses to forward data only to the intended recipient, improving efficiency and reducing unnecessary traffic.
- **Access Points (APs):** Enable wireless devices such as smartphones and laptops to connect to a wired network. They are critical components in Wi-Fi networks and must be secured to prevent unauthorized access.

II. CAREER OPPORTUNITIES IN NETWORK SECURITY

2.1. The Architect of Safety

Network Security Engineers play a crucial role in protecting organizational infrastructure. They design, implement, and maintain secure networks capable of resisting cyber threats such as Distributed Denial of Service (DDoS) attacks, intrusion attempts, and data breaches.

Their responsibilities include:

- Designing secure network architectures
- Configuring firewalls and intrusion detection systems
- Monitoring network traffic for anomalies
- Responding to security incidents

2.2. Emerging Roles

The field of network security continues to evolve, creating new specialized roles:

- **Network Security Administrator:**Manages firewalls, Virtual Private Networks (VPNs), and secure communication protocols.
- **Infrastructure Lead:**Oversees both physical and logical aspects of data center security, including redundancy and disaster recovery.
- **Intrusion Analyst:**Analyzes network traffic to detect patterns or “fingerprints” associated with cyberattacks.

2.3. Industry Demand

With the rapid development of technologies such as **5G networks** and the **Internet of Things (IoT)**, the number of connected devices is increasing exponentially. This expansion significantly enlarges the attack surface, making network security more critical than ever.

Companies—both in Vietnam (e.g., Da Nang tech hubs) and globally—are actively seeking professionals who can:

- Secure large-scale distributed systems
- Protect smart devices and IoT ecosystems
- Manage complex cloud-based infrastructures

III. THE OSI MODEL: THE UNIVERSAL LANGUAGE

3.1. Understanding the 7 Layers

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the functions of a networking system into seven distinct layers. It provides a structured approach to understanding how data flows across a network.

For cybersecurity professionals, the OSI model acts as a **diagnostic tool**, helping identify where vulnerabilities and attacks may occur.

3.2. Breakdown of the Layers

- **Layer 1 – Physical:**Handles physical transmission of data via cables, fiber optics, or wireless signals.
- **Layer 2 – Data Link:**Manages MAC addresses and switching within a local network.
- **Layer 3 – Network:**Responsible for logical addressing (IP addresses) and routing data between networks.

- **Layer 4 – Transport:** Ensures reliable or fast delivery of data using protocols such as TCP (reliable) or UDP (fast but less reliable).
- **Layer 5 – Session:** Establishes, manages, and terminates communication sessions between applications.
- **Layer 6 – Presentation:** Handles data encryption, compression, and formatting.
- **Layer 7 – Application:** Provides services directly to users through protocols such as HTTP, FTP, and SMTP.

3.3. Security Mapping

Each OSI layer is associated with specific types of attacks:

- **Layer 2 Attack:** ARP Spoofing
- **Layer 3 Attack:** IP Spoofing
- **Layer 7 Attack:** SQL Injection, Cross-Site Scripting (XSS)

Understanding this mapping helps security professionals implement targeted defenses at each layer.

IV. DEEP DIVE: PACKET ANALYSIS & WIRESHARK

4.1. What is a Packet?

In networking, data is not transmitted as a single continuous stream. Instead, it is divided into smaller units called **packets**. Each packet contains:

- **Header:** Metadata such as source and destination IP addresses
- **Payload:** The actual data being transmitted

This structure allows efficient routing and error handling across networks.

4.2. Tools of the Trade: Wireshark

Wireshark is a powerful **network protocol analyzer** used to capture and inspect network traffic in real time. It enables cybersecurity students and professionals to:

- Analyze packet-level data
- Detect suspicious activities
- Troubleshoot network issues

At university level (e.g., DTU), mastering Wireshark is essential for understanding real-world network behavior.

4.3. The TCP 3-Way Handshake

Before establishing a connection, TCP uses a three-step process:

1. **SYN:** The client initiates communication by sending a synchronization request.
2. **SYN-ACK:** The server acknowledges the request and responds.
3. **ACK:** The client confirms the connection, allowing data transfer to begin.

This handshake ensures reliable communication but can be exploited in attacks such as **SYN Flood**, where attackers overwhelm a server with incomplete requests.

4.4. Practical Lab

In practical sessions, students perform packet analysis using Wireshark:

- Capture real network traffic
- Identify unencrypted data transmissions
- Detect sensitive information such as passwords sent over HTTP

This exercise demonstrates why modern systems enforce **HTTPS encryption**, ensuring data confidentiality and integrity during transmission.